

Group Session with Feedback

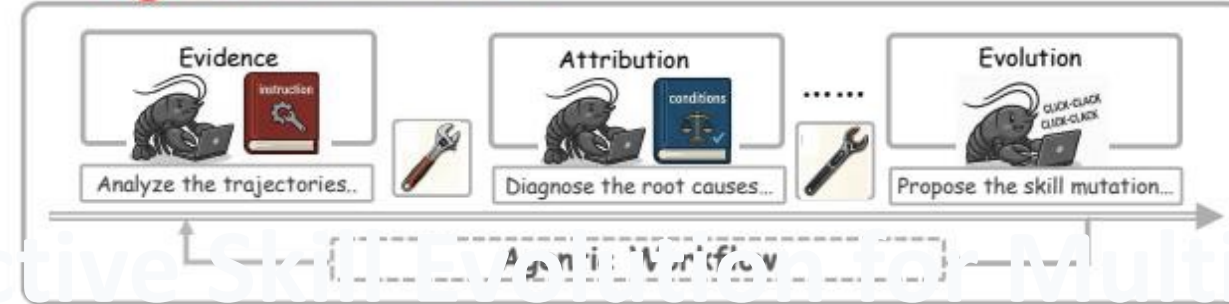


Agent Harness



Crab by Evolve Skill

Agentic Evolver



Evolve Existing skills

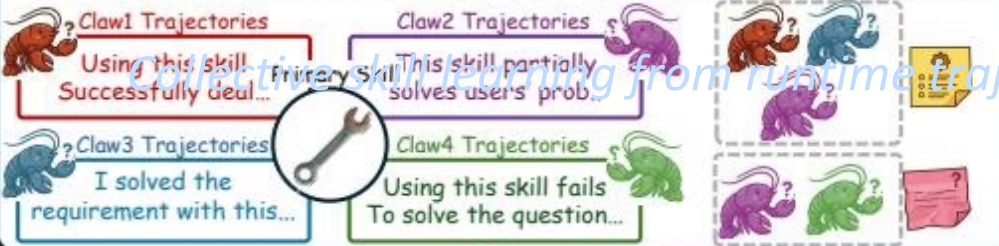


Create New Skills

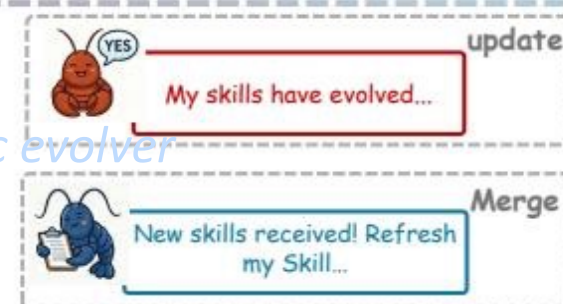


Pull

Group Real Sessions & Feedback from Env./Humans



Aggregation



Distribution of the Evolved Skill



Verification



ML researchers and AI systems engineers • arXiv:2604.08377

 Interaction In the Wild

Verification In the Wild 

Agent Skills Remain Static After Deployment

Static Paradigm

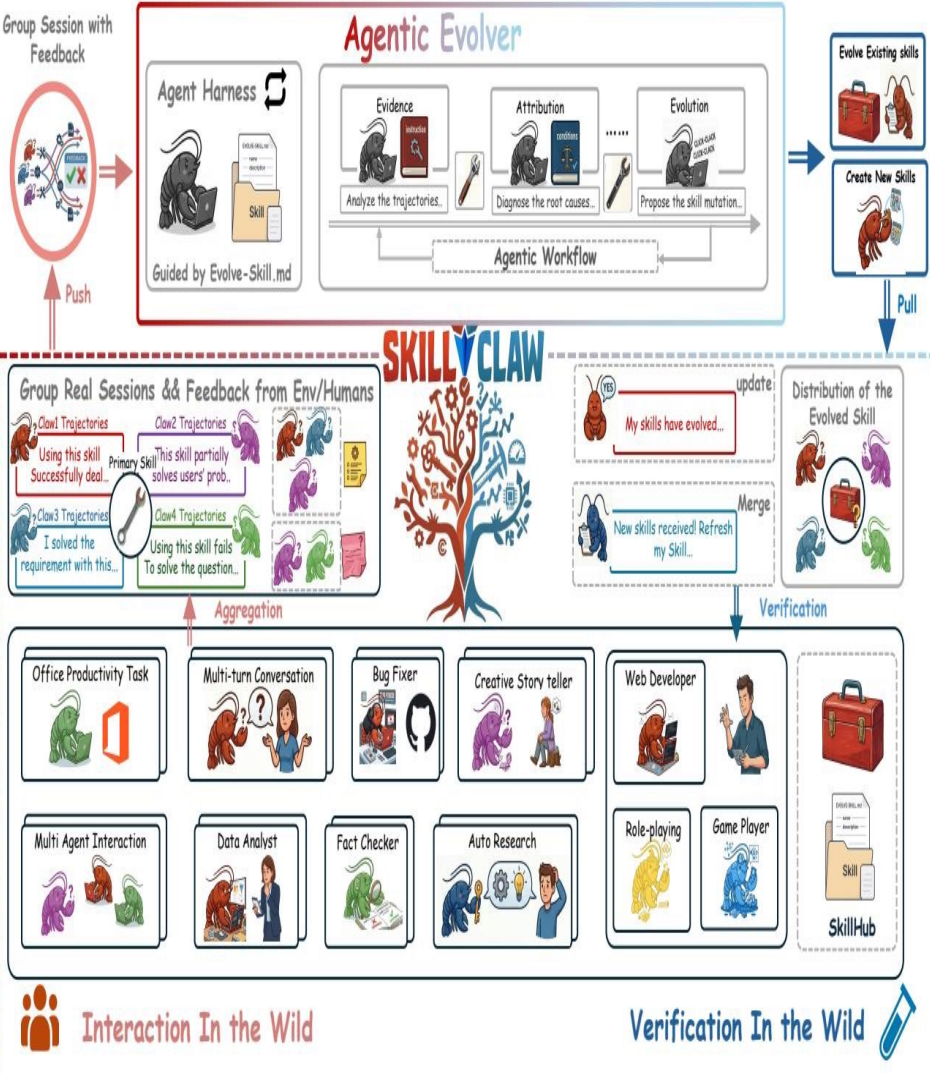
- Skill installed once
- Runtime failures repeat
- Users rediscover fixes independently
- No ecosystem-level memory

Evolving Paradigm (SkillClaw)

- Interactions become evidence
- Root causes are attributed
- Skills mutate and are validated
- Accepted updates synchronize globally

From repetitive rediscovery to shared improvement

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution



- | 1. Evidence | 2. Attribution | 3. Evolution | 4. Distribution |
|-----------------------------------|----------------------|--------------------------|---------------------------------|
| Aggregate cross-user trajectories | Diagnose root causes | Generate skill mutations | Validate + sync accepted skills |

Closed feedback loop from runtime use to ecosystem-wide updates

Agentic Evolver Pipeline: Evidence → Attribution → Evolution

Stage 1: Evidence Analysis

Collect trajectories across users and contexts; extract failure/success signatures tied to a skill.

Stage 2: Attribution

Reason over trajectories to identify root causes, boundary conditions, and missing behaviors.

Stage 3: Evolution

Propose mutated skill variants with open-ended agent reasoning, then pass to conservative validator.

Key property: evolution is agentic and open-ended, not rule-based patching.

Cross-User Aggregation Reveals Skill Behavioral Boundaries

Compatible Claw-style systems

OpenClaw CoPaw IronClaw
PicoClaw ZeroClaw NanoClaw
NemoClaw

Shared trajectory pool for one skill

- Diverse prompts, contexts, and environments
- Complementary success and failure traces
- Stronger evidence for generalizable updates

Why multi-user matters

A single user rarely generates enough signal to separate robust improvements from idiosyncratic fixes. Aggregation exposes true behavioral boundaries before mutation and validation.

Case Study: Multimodal Creative Pipeline Skill

Day-by-day evolution log (conservative verification; 1 accepted of 6)

Date	Outcome	Rationale
Day 1	Rejected	Overfits one prompt style; weak generalization across trajectories.
Day 2	Rejected	Improves ideation but reduces reliability in multimodal tool chaining.
Day 3	Accepted	Balances creativity and execution stability across diverse user sessions.
Day 4	Rejected	Regression on output consistency under longer context windows.
Day 5	Rejected	Adds verbosity without measurable trajectory-level benefit.
Day 6	Rejected	Narrow optimization; validator flags degradation on held-out traces.

Accepted 1 / 6 attempts

Case Study: Git Push Auth-Fallback Skill

Day-by-day evolution log (iterative refinement; 3 accepted of 6)

Date	Outcome	Rationale
Day 1	Accepted	Introduces resilient auth fallback path for common push failures.
Day 2	Accepted	Refines credential retry logic; improves success across user environments.
Day 3	Rejected	Change conflicts with secure defaults; validator rejects regression risk.
Day 4	Accepted	Adds clearer failure-state branching with stable behavior on traces.
Day 5	Rejected	No additional gain versus current best accepted variant.
Day 6	Rejected	Complexity increase without consistent trajectory-level improvement.

Accepted 3 / 6 attempts

Key Takeaways: What Distinguishes SkillClaw

Collective evolution

Individual interactions contribute to a shared, continuously improving skill ecosystem.

Fully automatic

Runtime interaction drives evolution without manual curation or explicit user intervention.

Agentic evolution paradigm

Open-ended reasoning over evidence generates context-aware skill mutations.

Implication: build agent ecosystems as continuously learning infrastructure, not static tool bundles.